

ECE 3337 — Mock Exam (Version 2)

Topics: Fourier Series + LTI Systems & Convolution + Laplace Transforms & Partial Fractions |
Total: 120 points | Time: 130 minutes

1. **(10 points)** Consider the sine/cosine Fourier series. Absolute integrability of a signal over one period is a condition required for the signal to be expandable in terms of sines and cosines. Show that absolute integrability of the signal over one period is sufficient to ensure the existence of the Fourier series coefficients as finite values.

You will need the triangle inequality:

$$\left| \int_a^b x(t) dt \right| \leq \int_a^b |x(t)| dt$$

2. **(5 points)** The Fourier coefficients of a periodic signal are $a_1 = 3e^{j\pi/4}$ and $a_{-1} = 3e^{-j\pi/4}$. Express the corresponding real signal in sine-cosine form. Assume the period is T_0 and the fundamental angular frequency is $\omega_0 = 2\pi/T_0$.

3. **(10 points)** A periodic signal is given as:

$$x(t) = 2 - |t|/\pi, \quad -\pi \leq t < \pi$$

Find the sine/cosine Fourier coefficients as well as the exponential Fourier series coefficients.

4. **(10 points)** Orthogonality: show that:

$$\int_{-T_0/2}^{T_0/2} \cos(m\omega_0 t) \cos(n\omega_0 t) dt = 0 \text{ for } m \neq n$$

Similarly show that $\int_{-T_0/2}^{T_0/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt = 0$ for $m \neq n$, and that $\int_{-T_0/2}^{T_0/2} \sin(n\omega_0 t) \cos(m\omega_0 t) dt = 0$ for all m, n . State the value each integral takes when $m = n$ (for the cosine-cosine and sine-sine cases).

5. **(5 points)** Calculate the output $y(t)$ of an LTI system whose impulse response is $h(t)$, when it is provided the input $x(t)$, where:

$$x(t) = u(t-3), \quad h(t) = e^{2-t}u(t)$$

6. **(5 points)** If the average power of a signal $x(t)$ is M , what is the average power of the signal $3x(4t)$?

7. **(10 points)** A signal $x(t)$ is zero for $t < -3$. It then rises linearly from 0 at $t = -3$ to a value of 6 at $t = -1$, stays constant at 6 from $t = -1$ to $t = 1$, decreases linearly back to 0 at $t = 4$, and is zero for $t > 4$. Write a formula for $x(t)$ in terms of unit step and unit ramp functions.

8. **(5 points)** Is the system $y(t) = (d/dt)[x(2 - t)]$ linear and time-invariant? Explain your reasoning.

9. **(5 points)** Consider an LTI system with input and output related through the equation:

$$y(t) = \int_{-\infty}^t e^{-(t-\tau)} x(\tau - 3) d\tau$$

What is the impulse response $h(t)$ for this system?

10. **(10 points)** The following are impulse responses of continuous-time LTI systems. Determine whether each system is causal and/or stable. Justify your answers.

$$(a) h(t) = e^{-5t} u(t - 1) \quad (b) h(t) = e^{-2t} u(4 - t)$$

11. **(10 points)** Consider the system with the input-output relationship:

$$y(t) = \int_{-\infty}^{\infty} f(t - \tau) x(\tau - 3) d\tau$$

Determine: (1) whether the system is Linear Time-Invariant (LTI); (2) if the system is LTI, find its impulse response $h(t)$ in terms of $f(t)$.

12. **(10 points)** Using the standard convolution results:

$$e^{at} u(t) * e^{bt} u(t) = [(e^{at} - e^{bt})/(a - b)] \cdot u(t), \quad a \neq b$$

compute the convolution $y(t) = e^{-3t} u(t) * e^{-5t} u(t)$ without evaluating any integrals. Then briefly explain how you would obtain the same result using Laplace transforms instead.

13. **(10 points)** Starting from the definition $X(s) = \int_0^{\infty} x(t) e^{-st} dt$, derive the Laplace transform of:

$$x(t) = e^{-at} \cos(\omega_0 t) u(t), \quad a > 0$$

State the region of convergence (ROC) associated with your result.

14. **(10 points)** Suppose $X(s)$ has only distinct, real poles, so that it can be written as:

$$X(s) = N(s)/[(s + p_1)(s + p_2) \dots (s + p_n)] = \sum_{k=1}^n A_k/(s + p_k)$$

Derive the general formula for the coefficient A_k by multiplying both sides by $(s + p_k)$ and evaluating at $s = -p_k$. Then use your formula to find the inverse Laplace transform $x(t)$ of:

$$X(s) = (9s + 14)/[(s + 2)(s + 7)]$$

15. **(10 points)** Now suppose $X(s)$ has a repeated real pole of order 2 at $s = -p$, together with a distinct real pole at $s = -q$:

$$X(s) = N(s)/[(s + p)^2(s + q)] = A_1/(s + p) + A_2/(s + p)^2 + B/(s + q)$$

Derive formulas for A_2 and B using the cover-up method, and derive the formula for A_1 by differentiating $(s + p)^2 X(s)$ with respect to s before evaluating at $s = -p$. Then use your formulas to find the inverse Laplace transform $x(t)$ of:

$$X(s) = (4s + 7)/[(s + 3)^2(s + 1)]$$